

HaloSight

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Chapter 1

Introduction

In the construction sector, where time and safety are paramount, real-time supervision and management of workforces is vital. This paper proposes the development of an innovative system based on video analytics and real-time personnel tracking, complemented by facial fatigue detection, specifically tailored for construction sites. By leveraging the insights from Imran et al. (2023) [2], this approach underscores the relevance of using deep learning and video analytics to enhance productivity and safety within the construction industry. The system aims to provide guidelines that alleviate operational inefficiencies and health risks while improving worker safety through the monitoring of worker movements, identification of work bottlenecks, and assessment of worker fatigue. In this way, the proposed method addresses the unique challenges of workforce management in construction, ultimately increasing both productivity and safety in modern-day construction sites.

1.1 Problem Statement

Workplace Health and Safety (WHS) concerns the welfare, safety, and physical and mental well-being of all employees in a workplace, while also addressing issues that protect the company from damage caused by workers. This system is designed to tackle the growing challenges faced by construction sites in enhancing workforce efficiency while maintaining safety. Many construction sites rely on traditional or partial solutions that fail to monitor the full-time dynamics of the workforce, missing critical opportunities to enhance operations. Additionally, a significant challenge is the inability to detect early signs of worker fatigue, a key determinant of workplace accidents and diminished performance among construction workers. Moreover, although executives in the construction industry claim to prioritize productivity, practical issues arise in productivity analysis and identifying the underlying reasons affecting performance. Consequently, construction sites waste resources, experience delays in project timelines, and jeopardize the safety of their workers. Not being able to anticipate safety hazards makes these

problems worse. It is evident that addressing and resolving these challenges involves more than just surface-level measures. Over time, this ignorant and insensitive behaviour leads to increased accidents, operational delays, and poor performance becoming common characteristics in the construction industry.

1.2 Scope

HaloSight is a real-time worker monitoring web application. It is a construction-place safety and compliance system. As a starter, it uses object detection to detect Personal Protective Equipment (PPE) violations to ensure worker safety.

HaloSight stands out from all other systems as it integrates the use of facial recognition to analyze stress among the workers to ensure productivity and avoid any mishaps. This system does not aim to be a general construction worker management system. Instead, it is focused on using object detection and facial recognition to increase productivity and safety of the workers on-site.

1.3 Modules

The application of HaloSight injects distinguished features to complete modules that will ensure it in becoming a comprehensive construction worker safety and compliance system. This documentation describes all of the system's

1.3.1 PPE Detection and Alerts

Ensuring safety compliance by monitoring personal protective equipment (PPE) usage through real-time video and picture analysis.

1. **Feature 1** (Supervisor): Real-time video analytics for PPE detection using object detection.
2. **Feature 2** (Supervisor): Supervisors receive immediate non-compliance alerts for quick action.
3. **Feature 3** (Worker): Alerts for non-compliance to the staff, reminding workers to wear or adjust their required Personal Protective Equipment.

1.3.2 Facial Emotion and Stress Analysis

Monitors worker emotional states to promote well-being and productivity by using facial recognition and emotional analysis.

1. **Feature 1** (Supervisor): Emotion detection through facial analysis to monitor stress, fatigue, and discomfort in workers.
2. **Feature 2** (Supervisor): Automatic notifications are sent to supervisors when elevated stress levels are detected.
3. **Feature 3** (Worker): Workers can view their own emotional status and receive feedback on stress or fatigue.

1.3.3 Reward and Recognition System

Encouraging workers to follow safety protocols by offering rewards for consistent safety compliance.

1. **Feature 1** (Supervisor): Tracks workers who consistently follow safety regulations and protocols.
2. **Feature 2** (Supervisor): Manage the reward system and distribute rewards (e.g. bonuses, perks) to workers who maintain high safety standards.
3. **Feature 3** (Worker): Workers can view their compliance progress and eligibility for rewards.

1.3.4 Data History and Report Generation

Monitors safety and emotional trends over time to provide insights for better decision-making and improving workplace practices.

1. **Feature 1** (Supervisor): Collection and storage of historical data related to worker safety and emotional states.
2. **Feature 2** (Supervisor): Automated report generation highlighting trends, including stress levels and PPE compliance.
3. **Feature 3** (Worker): Workers can view their personal historical data related to safety and emotional well-being.

1.3.5 Web Interface Development

Focuses on creating a user-friendly web interface that ensures easy interaction and accessibility for both workers and administrators.

1. **Feature 1** (Supervisor and Worker): Development of a user-friendly web application interface, designed for optimal usability and efficient workflow navigation.
2. **Feature 2** (Supervisor and Worker): Responsive design to ensure compatibility across all devices, allowing access from desktops, tablets, and mobile devices.

1.3.6 System Integration and Performance

Unifies all system components (PPE detection, emotional analysis, alerts) to ensure smooth, integrated workflows and real-time functionality.

1. **Feature 1** (Supervisor and Worker): Integration of all the different components into one system.
2. **Feature 2** (Supervisor and Worker): Performance optimization to ensure fast, real-time monitoring and smooth operation with low latency.

1.4 User Classes and Characteristics

Table 1.1: User Classes

User class	Description
Admins/ Supervisors	Supervisors are responsible for overseeing daily operations on-site, ensuring that workers adhere to safety protocols and regulations. They receive real-time alerts on PPE non-compliance and monitor emotional health indicators such as stress and fatigue levels. Supervisors also manage and distribute rewards through the recognition system, tracking worker performance and compliance. Additionally, they are responsible for generating detailed safety and productivity reports, coordinating with other stakeholders, and taking necessary actions to improve workplace safety and overall efficiency.
Construction Workers	Construction workers are the employees that perform on site tasks while adhering to safety protocols. Workers receive real-time alerts for any PPE non-compliance or safety issues, prompting immediate corrective actions. They have access to their personal emotional health data, allowing them to monitor stress, fatigue, and overall well-being. Workers also participate in the reward system, tracking their safety compliance and performance to earn recognition and incentives. By following system feedback, they can continually improve their safety practices and contribute to a safer, more productive work environment.
Construction Company	The construction company that deploys the software, has access to monitor safety compliance reports, emotional health analytics, and safety metrics for workers on-site. They can view historical data, generate reports, and track overall system performance to ensure operational standards. Additionally, they manage administrative functions such as granting user access and reviewing system performance metrics, without direct responsibility for on-site management.

Chapter 2

Project Requirements

This chapter defines and discusses both the functional and non-functional requirements required for proper design, implementation and functionality of the project. It describes the processes that the system should support, and the quality of service requirements that guarantee dependability, interoperability, and extensibility. It is against these requirements that the framework for the development of a strong and effective system is established to meet the needs of all the stakeholders.

2.1 Use-case Diagram

The Use Case Diagram gives an overview of how the user and other components of the HaloSight system will interact with the different functions in the system. Here, the key aspects have been described, including PPE detection, stress level detection, reporting, and reward functionality. A use case describes the goal of the system, whereas the actors, which may be the workers, supervisors, or the system itself, show how these stakeholders employ the functionalities of the system to achieve that goal. This diagram helps as a map to explain how various users interact with the system to promote safety, compliance, and productivity in a workplace.



Figure 2.1: Use Case Diagram for HaloSight

2.2 Extended Use Cases

Use Case 01: Detect PPE Compliance

Scope: HaloSight System (Worker Safety and Compliance Monitoring)

Level: User-goal

Primary Actor: System (HaloSight)

Stakeholders and Interests:

- **Worker:** Wants to ensure compliance with PPE rules to maintain safety.
- **Supervisor:** Needs to ensure that workers are adhering to safety regulations and receive notifications for non-compliance.
- **Admin:** Needs the system to be reliable and provide accurate compliance data for all workers.

Preconditions:

- Workers are on-site, and the system is actively monitoring the workplace.
- The video feed from the site is functioning and accessible by the HaloSight system.

Postconditions:

- The system has detected whether the worker is in compliance with PPE requirements.
- If non-compliance is detected, the system sends alerts to both the worker and the supervisor.

Main Success Scenario:

Worker Action	System Response
1. Worker enters the worksite.	
	2. System begins monitoring worker's PPE through video feed.
	3. System analyzes the worker for the required PPE (e.g., helmet, gloves).
	4. System detects that the worker is fully compliant with PPE requirements.
	5. System logs compliance data for that worker.

Extensions:

- **3a.** Worker enters the worksite without wearing the required PPE.
System Response: The system detects a PPE violation and immediately generates an alert.
- **3b.** The system encounters an error in the video feed.
System Response: The system logs the issue and sends a notification to the admin for resolution.

2.3 Functional Requirements

Below are the functional requirements for the HaloSight system, designed to ensure comprehensive monitoring and safety compliance for construction workers. The system is divided

into distinct modules, each with specific functionalities. These requirements define the key capabilities of each module.

2.3.1 Module 1: PPE Detection and Alerts

The following are the functional requirements for the **PPE Detection and Alerts** module:

1. The system shall use real-time video analytics to detect if a worker is missing any required PPE (e.g., helmets, gloves, etc.).
2. The system shall alert the supervisor immediately if a PPE violation is detected.
3. The system shall send a notification to the worker in case of a PPE violation.
4. The system shall display real-time PPE compliance status on the supervisor's dashboard for all workers.
5. The system shall maintain a historical log of all PPE violations and corresponding actions taken.
6. The system shall allow the supervisor to generate reports on PPE compliance trends over time.

2.3.2 Module 2: Facial Emotion and Stress Analysis

The following are the functional requirements for the **Facial Emotion and Stress Analysis** module:

1. The system shall analyze workers' facial expressions in real-time to detect signs of stress, fatigue, and discomfort.
2. The system shall send an automatic alert to the supervisor if high stress or fatigue levels are detected in a worker.
3. The system shall display the emotional state of workers on the supervisor's dashboard.
4. The system shall provide workers with access to their own emotional state, including stress or fatigue levels, through a personal dashboard.
5. The system shall store emotional state data in a historical log for trend analysis.
6. The system shall notify workers if their emotional state indicates high levels of stress or fatigue.

2.3.3 Module 3: Reward and Recognition System

The following are the functional requirements for the **Reward and Recognition System** module:

1. The system shall track each worker's compliance with safety protocols, including PPE usage and emotional well-being.
2. The system shall assign reward points to workers who consistently follow safety protocols.
3. The system shall allow supervisors to manage and distribute rewards (e.g., bonuses or perks) based on the worker's compliance status.
4. The system shall provide workers with access to their compliance score and reward status through a personal dashboard.

2.3.4 Module 4: Data History and Report Generation

The following are the functional requirements for the **Data History and Report Generation** module:

1. The system shall collect and store historical data related to worker safety and emotional states.
2. The system shall automatically generate reports highlighting trends in PPE compliance and emotional health.
3. The system shall allow supervisors to view and customize historical data reports based on time periods, worker compliance, and safety trends.
4. The system shall allow workers to view their personal historical data related to safety and emotional well-being.

2.3.5 Module 5: Web Interface Development

The following are the functional requirements for the **Web Interface Development** module:

1. The system shall provide a user-friendly web application interface for both workers and administrators to interact with the system.
2. The system shall ensure a responsive design that works seamlessly across desktops, tablets, and mobile devices.

3. The system shall allow users (both workers and administrators) to access the system from any location via a web interface.
4. The system shall allow real-time updates on worker compliance, emotional health, and alerts through the web interface.

2.3.6 Module 6: System Integration and Performance

The following are the functional requirements for the **System Integration and Performance** module:

1. The system shall integrate all components, including PPE detection, emotional analysis, alerts, and report generation, into one unified platform.
2. The system shall ensure that all integrated components function in real-time, providing immediate feedback and updates.
3. The system shall be optimized for performance, ensuring low latency in real-time video processing, notifications, and data updates.
4. The system shall support scalability, allowing multiple construction sites and numerous workers to be monitored without performance degradation.

2.4 Non-Functional Requirements

2.4.1 Reliability

1. The system shall operate with no less than 1,000 Mean Time Between Failure (MTBF) operational hours.
2. The system shall guarantee that failure in one module (e.g., PPE detection) does not affect the performance of other modules (e.g., facial emotion analysis).
3. The system shall log failures and crashes automatically without user intervention, and the error recovery strategy shall enable the system to resume working within 1 minute of failure detection.

2.4.2 Usability

1. The intended web interface of the system shall enable new users to accomplish simple operations such as automatically reporting their compliance status within five minutes of

first use, with no training required.

2. The system shall be designed with a user-friendly interface that minimizes errors and allows users to recover from errors within three clicks.
3. The system shall be available to workers and supervisors via mobile devices, tablets, and desktops with an adaptive layout that conforms to Web Content Accessibility Guidelines (WCAG) 2.1 standards.
4. The system shall ensure that critical actions (e.g., acknowledging a PPE violation) can be performed in two or fewer mouse clicks.

2.4.3 Performance

1. The system shall process PPE detection and facial emotion recognition from video feeds with an input-to-alert generation time of less than 2 seconds.
2. The system shall support up to 1,000 concurrent users (including managers and workers) without noticeable performance degradation.
3. The system shall generate a report on PPE compliance or emotional health trends for the past month in less than 5 seconds.
4. The system shall be capable of streaming data from multiple construction sites simultaneously with less than 3% downtime over a period of 1 year.

2.4.4 Security

1. The system shall encrypt all user data, including video feeds, using AES-256 encryption.
2. Admin accounts shall require multi-factor authentication for access to sensitive data.
3. The system shall log all user activities, including access to personal dashboards and report generation, with logs retained for 6 months for auditing purposes.
4. The system shall be resistant to security breaches, with an estimated Time-to-Break (TTB) of at least 6 hours for a skilled White Hat hacker.
5. The system shall lock accounts after three failed login attempts and send a reset link to the registered email address.

2.4.5 Availability

1. The system shall have an availability of 99.9%, allowing for a maximum of 8.76 hours of downtime per year.
2. The system shall ensure that all critical services (e.g., PPE detection, alert generation) are accessible during working hours with no more than 1 minute of downtime.

2.4.6 Scalability

1. The system shall scale to accommodate a 50% increase in workers and construction sites without requiring major modifications to the underlying infrastructure.
2. The system shall be capable of handling data from up to 50 construction sites simultaneously while maintaining performance within acceptable limits.

2.4.7 Maintainability

1. The system shall allow updates to components (e.g., the PPE detection module) while the system is operational, with updates taking no longer than 10 minutes.
2. The system shall be modular, allowing individual components (e.g., facial recognition or report generation) to be updated without impacting the functionality of other modules.

2.4.8 Compliance

1. The system shall comply with OSHA (Occupational Safety and Health Administration) regulations and requirements for occupational health and safety systems.
2. The system shall comply with GDPR (General Data Protection Regulation) standards for data storage, access, and deletion in relation to user privacy.

2.5 Domain Model

The following is the domain model of the HaloSight project which shows how entities such as workers, supervisors, emotional health systems, PPE detection systems etc. are associated with each other. It shows how the system records the emotions and safety data and handles compliance while producing alerts and reports on workplace safety and performance.

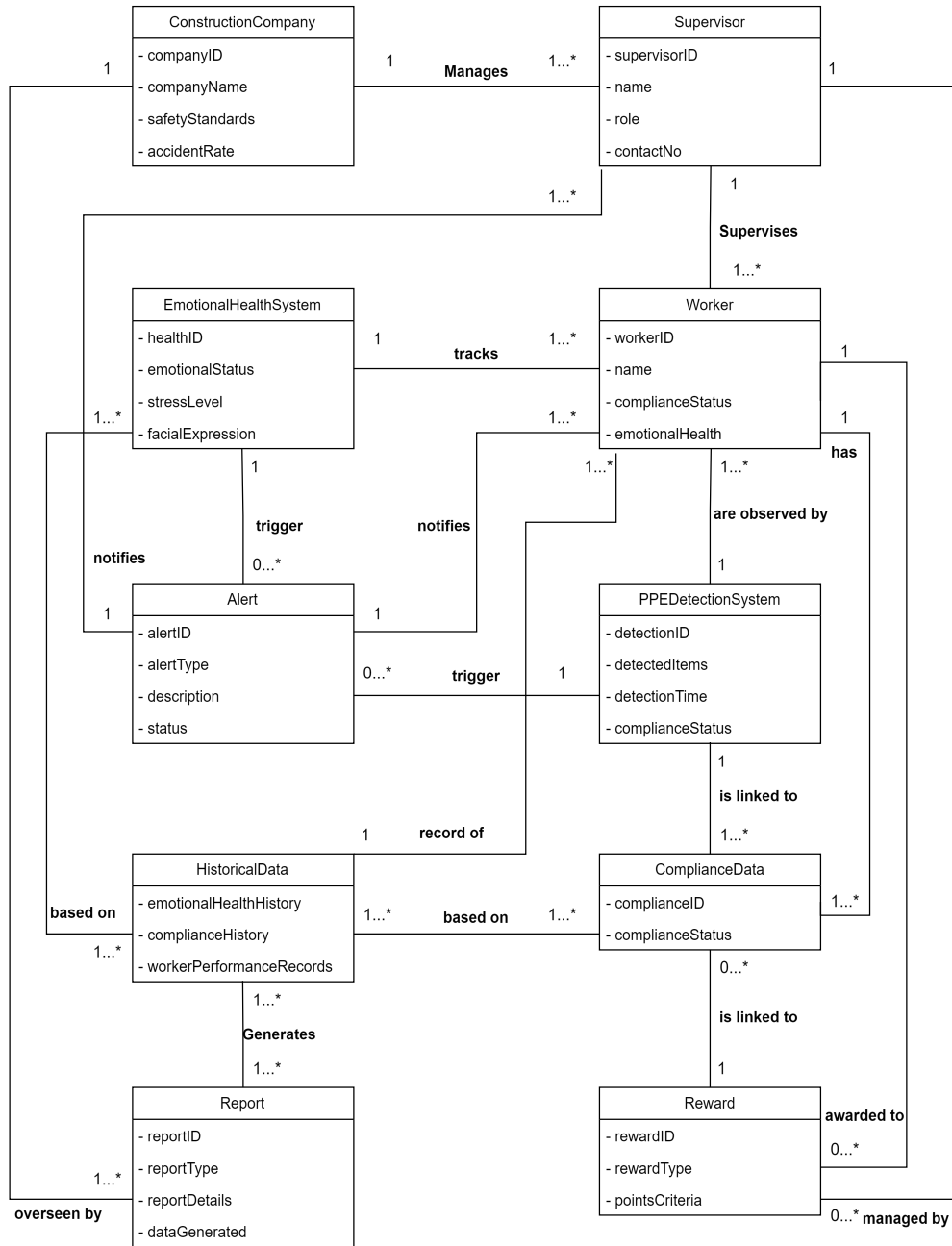


Figure 2.2: Domain Model for HaloSight

Chapter 3

System Overview

HaloSight is a holistic worker safety surveillance tool developed for construction sectors. The first objective of the system is the real-time identification of safety protocols' infringements, including the incorrect application of the Personal Protective Equipment (PPE), and the second one is the worker's condition assessment by facial emotions and stress levels. As a result of using video analytics and artificial intelligence, HaloSight provides supervisors with notifications in cases of non-adherence to PPE and signs of worker drowsiness.

The features of the system include; identification of PPE, real-time alerts, historical records, and generation of reports to monitor the workers' compliance and their emotional well-being. Also, it organizes and encourages positive safety behavior through a reward and recognition program. The system helps organizations to adhere to safety standards these include OSHA thus offering businesses a means of improving on the safety and productivity of the workers.

3.1 Architectural Design

The system follows the **Client-Server System**, with microservices implemented at the server end, where the client side includes the supervisor and worker interfaces and the server end handles video data, PPE, ED, and reporting. All are the modules incorporated into this architecture so that communication between client and server can take place easily. Seperate components are listed and defined below:

3.1.1 Client-Side or Front-End

The Client-Side involves two interfaces: **Supervisor Interface** and **Worker Interface**.

Description:

The client layer is composed of graphical user interfaces (GUI) used by users with the roles of Supervisor and Worker. All interfaces are accessible through web or mobile applications.

Functionality:

- **Workers:**
 - Receive live notifications about PPE compliance.
 - Check personal compliance status and records.
- **Supervisors:**
 - Supervise the conformance level of workers and monitor their emotional condition.
 - Receive notifications for PPE violations or high stress levels.
 - Generate compliance and emotional health reports.

3.1.2 Server-Side (Microservices)

The server-side is divided into microservices, each responsible for a particular system functionality. These microservices use lightweight protocols such as RESTful APIs and gRPC to support real-time data flow and enable scaling.

3.1.2.1 PPE Detection Service

- **Description:** This microservice processes video streams to identify the level of PPE usage in real-time.
- **Functionality:**
 - Captures real-time video feeds from work sites.
 - Uses computer vision algorithms to recognize personal protective equipment (e.g., helmets, gloves).
 - Sends alerts to the worker and the supervisor in case of PPE violations.

3.1.2.2 Emotion and Stress Analysis Service

- **Description:** This service tracks the facial expressions of workers to determine their emotional state and stress levels.

- **Functionality:**

- Processes real-time video through emotion recognition algorithms.
- Detects symptoms of fatigue, pressure, or discomfort.
- Notifies supervisors if the worker's stress level is high.

3.1.2.3 Compliance and Reward Management Service

- **Description:** In this service, the firm monitors the extent to which its workers are observing the safety measures and administers the incentive program on the same.

- **Functionality:**

- Records compliance and violation information from the PPE detection and emotion analysis services.
- Monitors compliance progress in the long run and provides incentives to the employees.
- Enables supervisors to set rules governing rewards and control incentives.

3.1.2.4 Data Storage and Reporting Service

- **Description:** It is responsible for the retention of compliance and emotional health data for a longer period and generation of reports.
- **Functionality:**
 - Saves the violation records of PPE and the trend of emotion in a relational database (such as MySQL or PostgreSQL).
 - Prepares reports on the level of compliance of the workers and their emotional state to be discussed with managers.
 - Includes an API for history information retrieval as well as on-demand report generation.

3.1.2.5 Real-Time Alerts and Notification Service

- **Description:** This microservice is in charge of all real-time notification and alerting.
- **Functionality:**

- Informs workers and supervisors instantly in situations when PPE is not used correctly or when stress levels are high.
- Administers both push notifications and email and/or SMS notifications for important announcements.

3.1.2.6 Authentication and User Management Service

- **Description:** Controls the user authentication and it is responsible for granting access to the system.
- **Functionality:**
 - Identifies the workers, supervisors, and administrators.
 - Offers the concept of Role-Based Access Control (RBAC) to guarantee that every user in the system has the rights that they should possess.

The following is the architectural diagram for the HaloSight System:

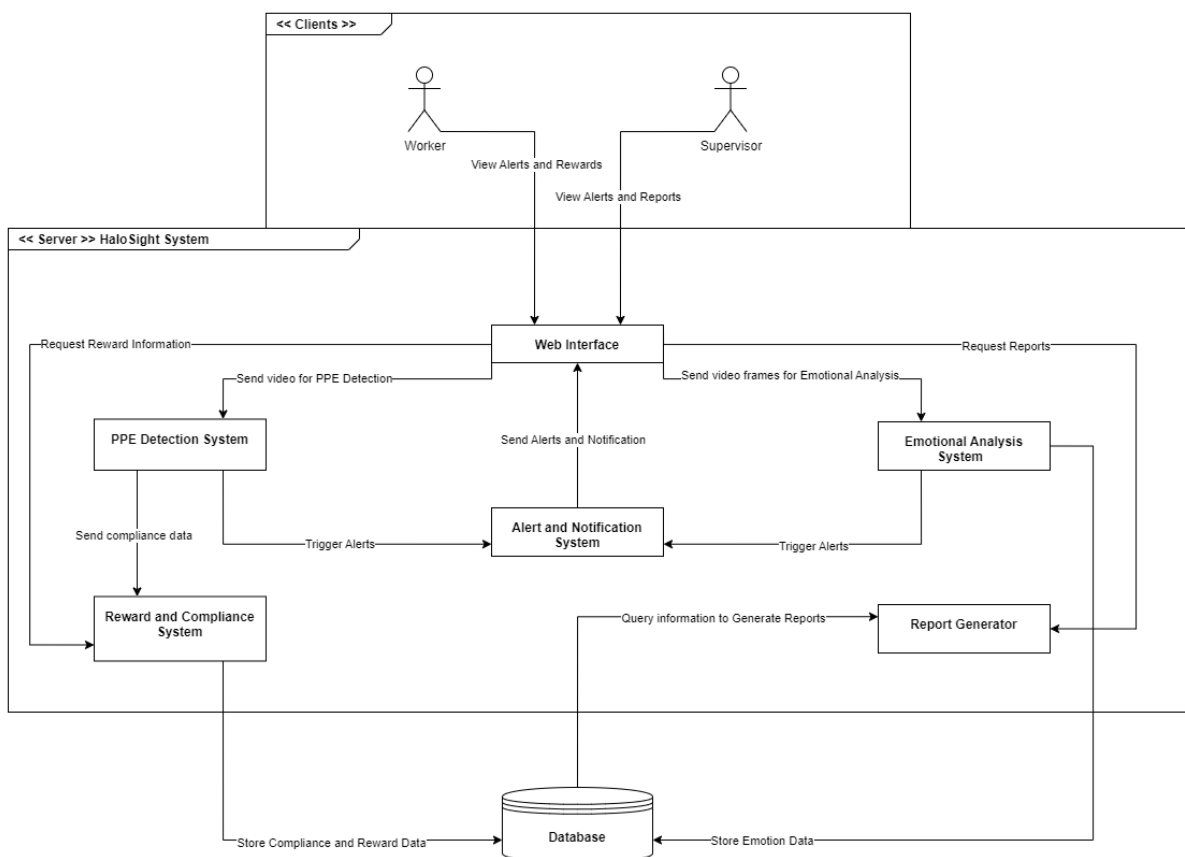


Figure 3.1: Client-Server architectural diagram

3.2 Design Models

In this system, we will follow a procedural approach, where each process follows a defined sequence of steps for data input, analysis, and response. The system operates in a linear manner, handling PPE detection and alerts systematically from input to output. This allows clear control over data flow and process execution at each step..

3.2.1 Activity Diagram

The activity diagram belongs to "UC 01: Detect PPE Compliance", and it starts with the worker entering the worksite. The system checks whether the worker is wearing the required PPE by analyzing video input. If the worker is detected without PPE, the system sends an alert to the supervisor, as well as the respective worker, to correct their PPE compliance status. Upon receiving the alert, the supervisor reviews the situation and takes appropriate action, such as addressing the violation. If the worker is compliant with the PPE rules, the system logs the compliance and no further action is taken. The system then may continue to monitor other workers.

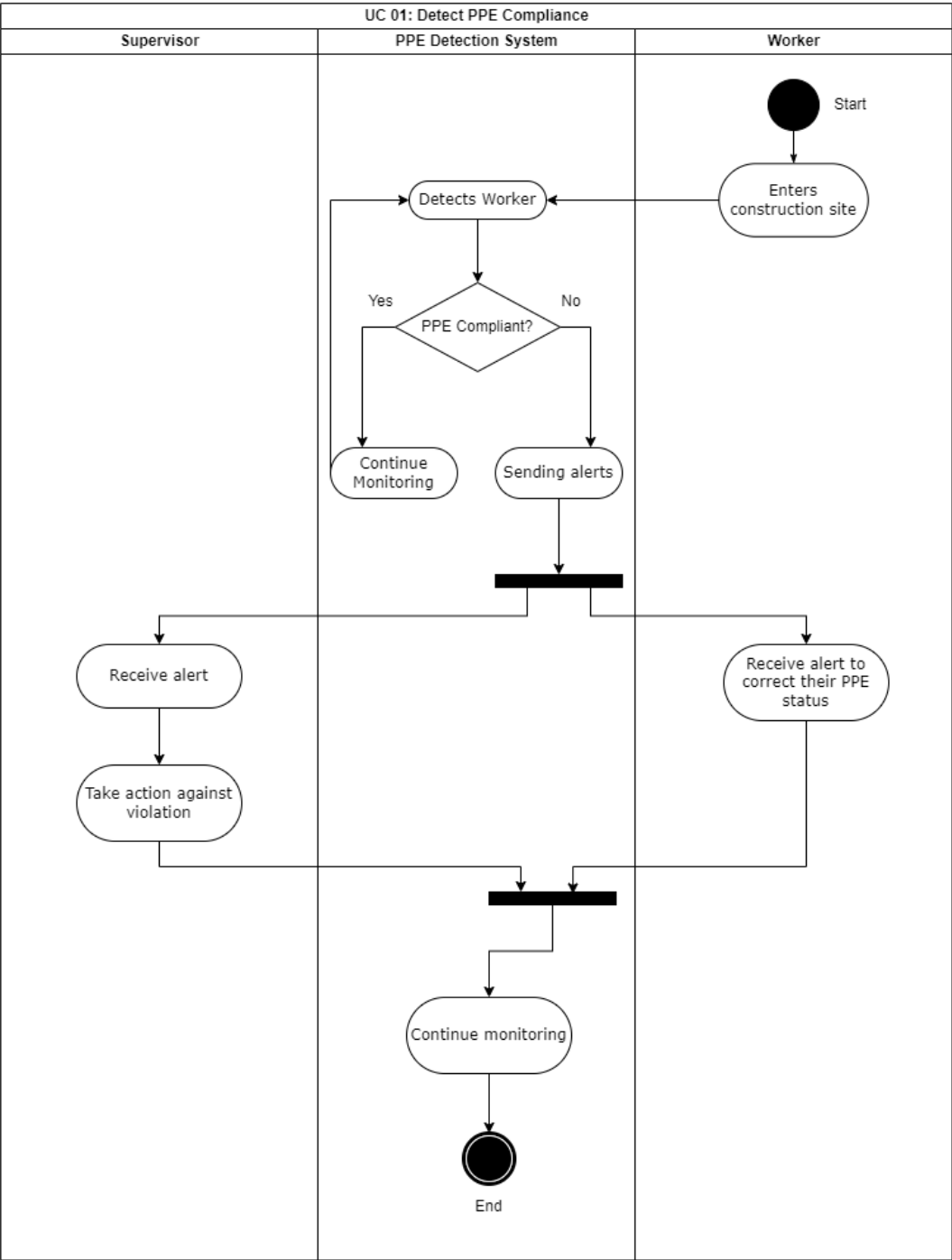


Figure 3.2: Activity Diagram for UC:01 Detect PPE Compliance

3.2.2 Data Flow Diagram

The Level 1 Data Flow Diagram for "UC 01: Detect PPE Compliance" provides an overall view of the PPE detection system. The "PPE Detection System" processes the worker's data, analyzing their compliance with PPE requirements. If the worker is not compliant, the system generates alerts and sends to the worker and to the Supervisor as a sink. Both compliant and non-compliant data are stored in the "Compliance Data" store for future reporting and analysis, the compliance events are logged in "Detection Logs" store, while the workers have their information stored in the "Worker Database".

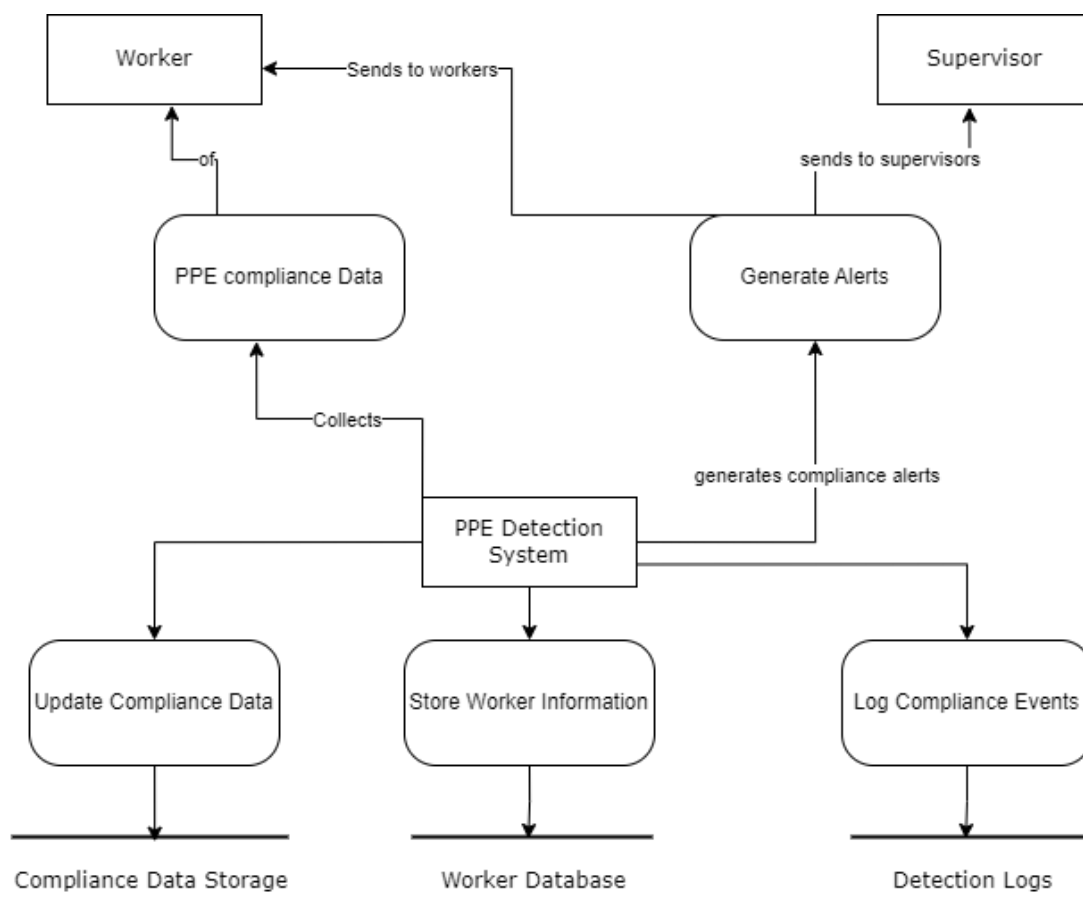


Figure 3.3: Level 1 DFD for UC 01: Detect PPE Compliance

The Level 2 Data Flow Diagram focuses on the detailed process within the PPE detection system. It breaks down the process into smaller steps which include the use of video processing and analytics to check for compliance. If a violation is detected, an alerts are sent. If compliance is confirmed, the data is logged into the "Compliance Data" store. The "Detection Logs" store keeps records of both violations and compliance for further use, like generating reports or analyzing safety trends.

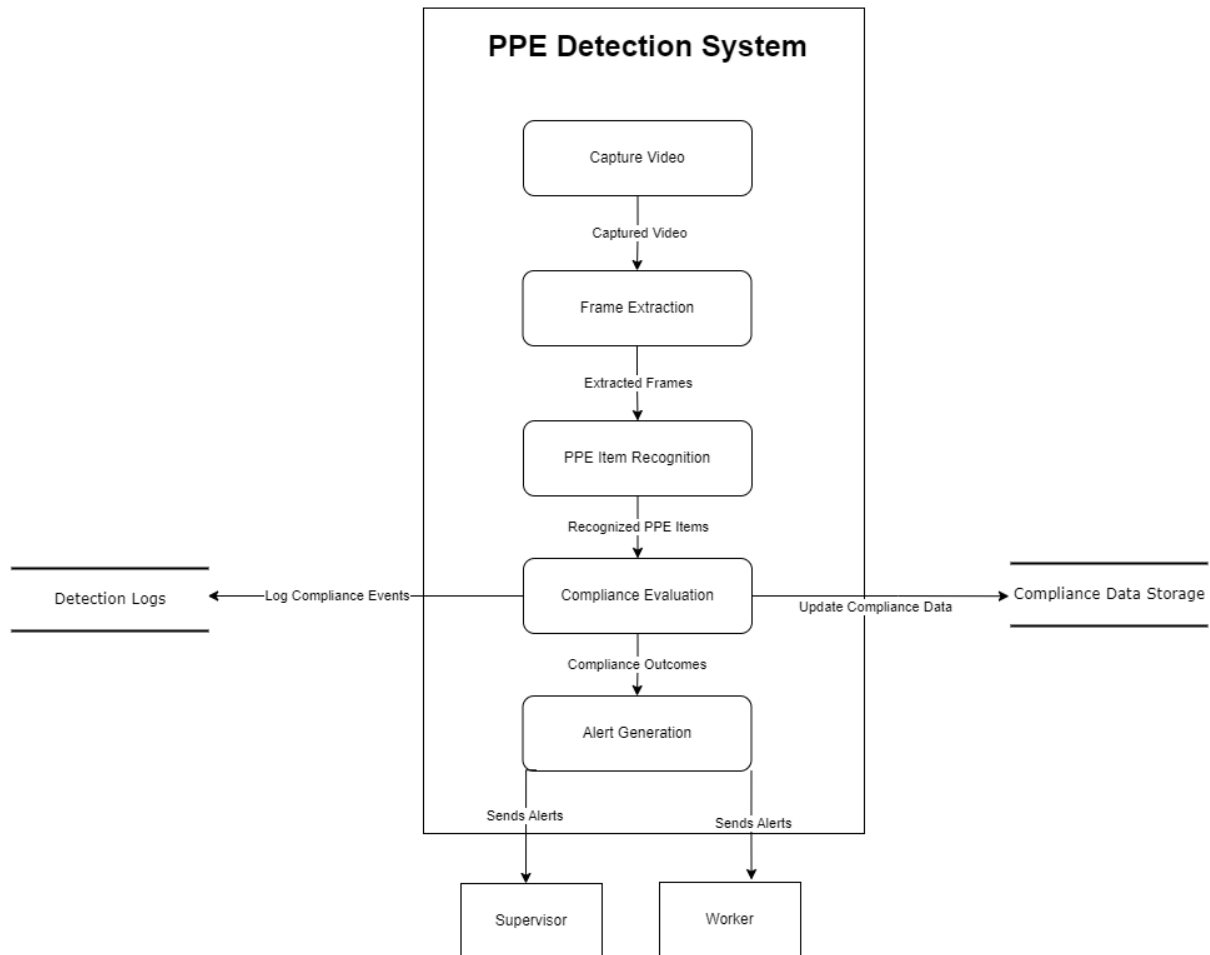


Figure 3.4: Level 2 DFD for UC 01: Detect PPE Compliance

3.2.3 System-Level Sequence Diagram

The system sequence diagram represents the process flow for the PPE Detection use case. It clearly outlines these interactions between the worker, PPE detection system, and supervisor, showing how the system responds to various inputs.

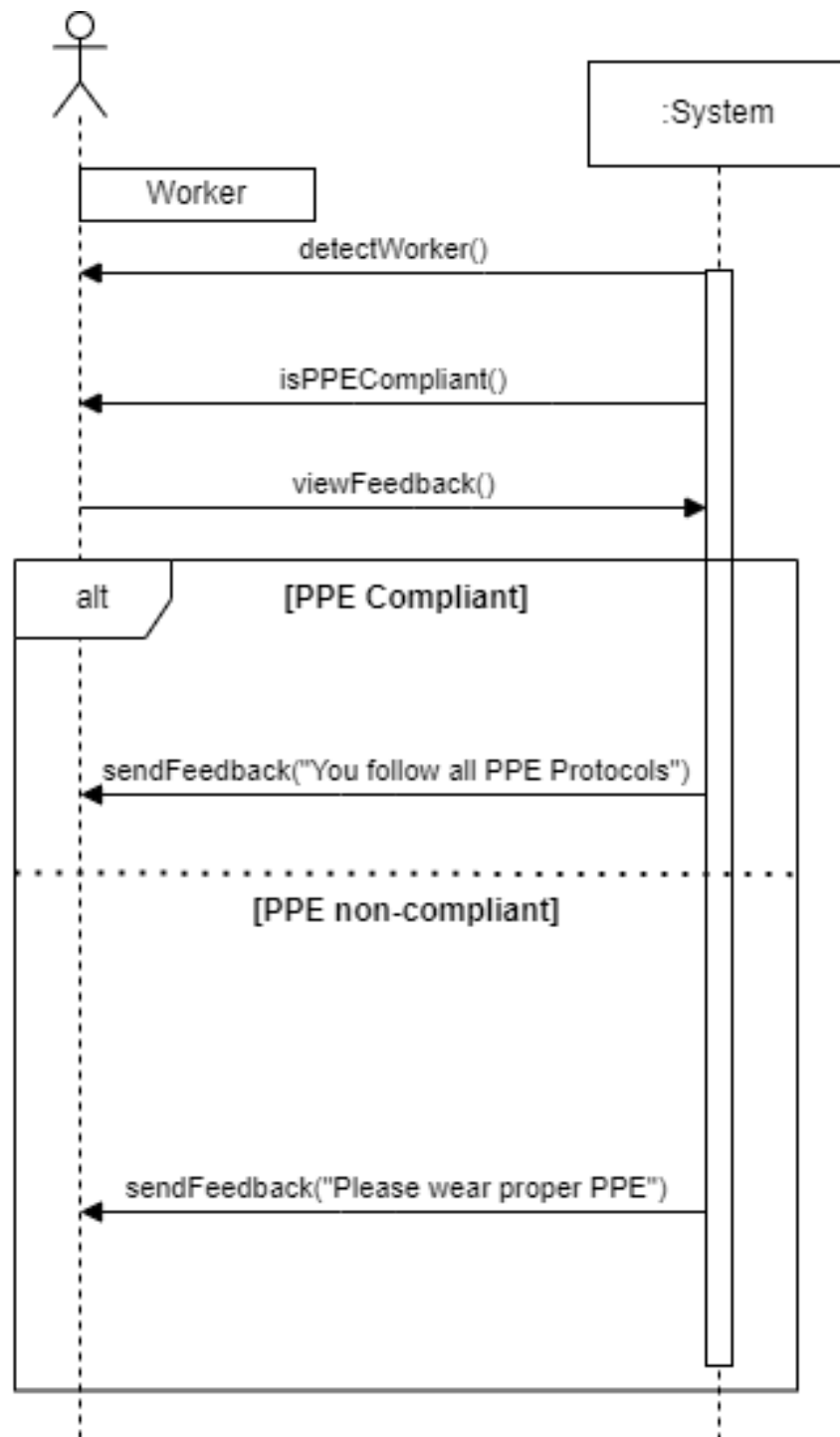


Figure 3.5: System Sequence Diagram for UC 01: Detect PPE Compliance

3.2.4 State Transition Diagram

The state transition diagram for the PPE Detection System illustrates how the system moves through various states in response to detecting PPE compliance. Initially, the system remains in an idle state, waiting for an event where a worker enters the work-site. Upon detecting a worker's entry, the system transitions to the "Detecting PPE" state, where video analytics are used to assess whether the worker is compliant with PPE requirements. If the worker is compliant, the system moves to the "PPE Confirmed" state and then returns to the idle state. However, if a violation is detected, the system transitions to the "Violation Detected" state and triggers alerts. In the alerting phase, notifications are sent to both the supervisor and the worker. Once these alerts are processed, the system returns to the idle state, ready for the next event.

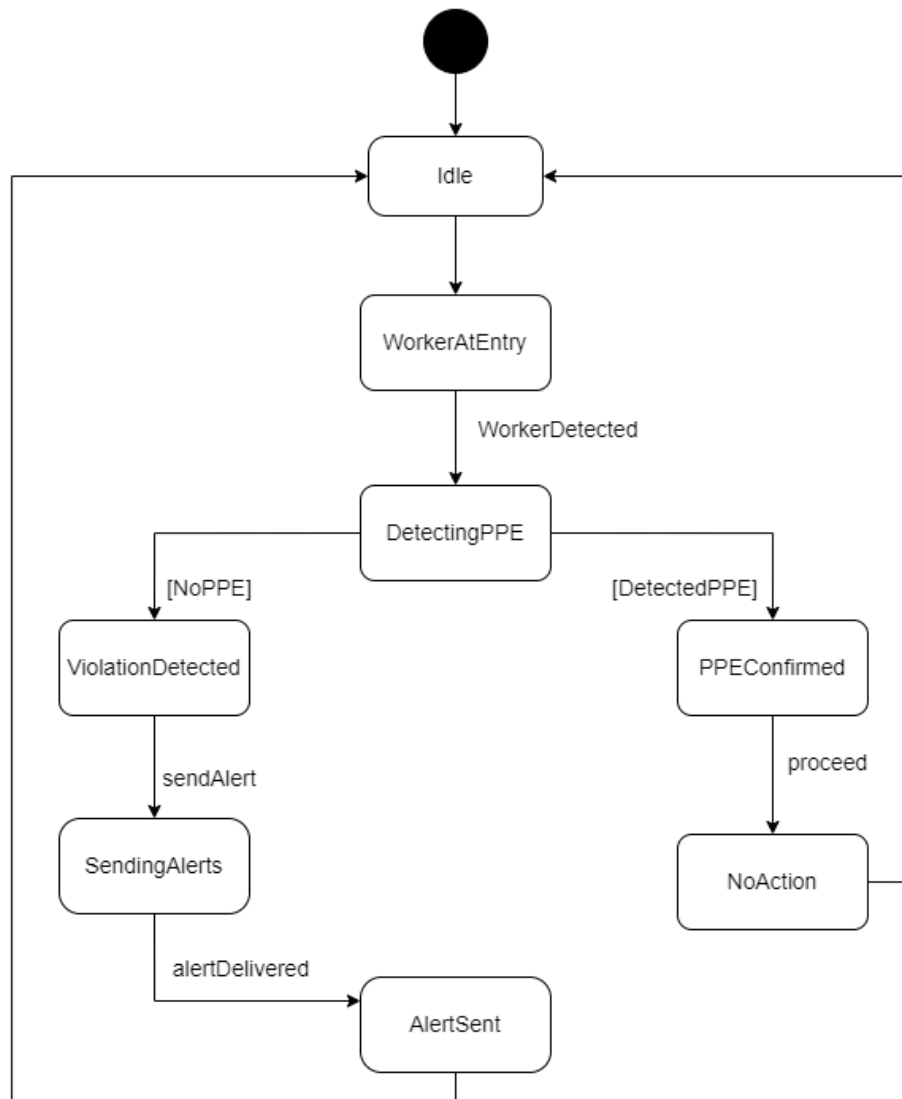


Figure 3.6: State Transition Diagram for UC 01: Detect PPE Compliance

3.3 Data Design

3.3.1 Selecting Unstructured Databases

In our project, we are utilizing unstructured and semi-structured databases such as MongoDB and PostgreSQL due to their inherent flexible qualities. These databases are particularly well-suited for our needs because:

Flexibility: Unlike traditional relational databases that require a predefined schema, unstructured databases allow us to store data in various formats. This is beneficial for our project as we are dealing with different types of data (like video files, compliance records, and emotional analysis data) that may evolve over time without requiring extensive schema modifications.

Scalability: Unstructured databases can easily accommodate large volumes of data, making them ideal for our PPE Detection System, which may generate substantial data from video feeds and compliance records as it scales.

Rich Data Types: These databases support complex data types such as arrays and nested documents, enabling us to represent relationships and hierarchies more naturally. For instance, a worker's compliance history can include multiple records without needing separate tables.

Performance: By using document-based storage, we can retrieve entire documents in a single query, which improves performance, especially when dealing with large datasets, such as those generated by the video analysis for PPE compliance.

3.3.2 Information Transformation into Data Structures:

For unstructured databases, it is considerably harder to visually display the schema, due to their abstract nature and scalable structure. However, despite these limitations, we provide a visual representation of our schema to offer a clearer understanding of the relationships and entities involved in the system.

We will understand the data structures better with the example of our Use Case 01: "Detect PPE Compliance". The information domain of our PPE Detection System is transformed into data structures through a combination of entity definitions and the relationships between them. The major data entities include:

PPEComplianceRecord: This entity stores compliance records for each worker, including a unique record ID, date of compliance check, status, and associated video data.

Worker: Represents each worker with attributes such as worker ID, name, role, contact details, compliance history, and reward points.

Supervisor: Contains information about supervisors who oversee the workers, including their ID, name, and contact details.

Alert: Represents alerts generated when violations are detected, linking back to the worker involved.

Reward: Stores information about rewards earned by workers for compliance.

HistoricalData: Maintains historical records of compliance checks and emotional analysis over time for each worker.

These entities can be stored as documents within a MongoDB collection, where each document can contain all relevant information for that entity in a single structure. This design allows for easy updates and retrievals, streamlining the data management process. The following is a visual representation of the JSON schema:

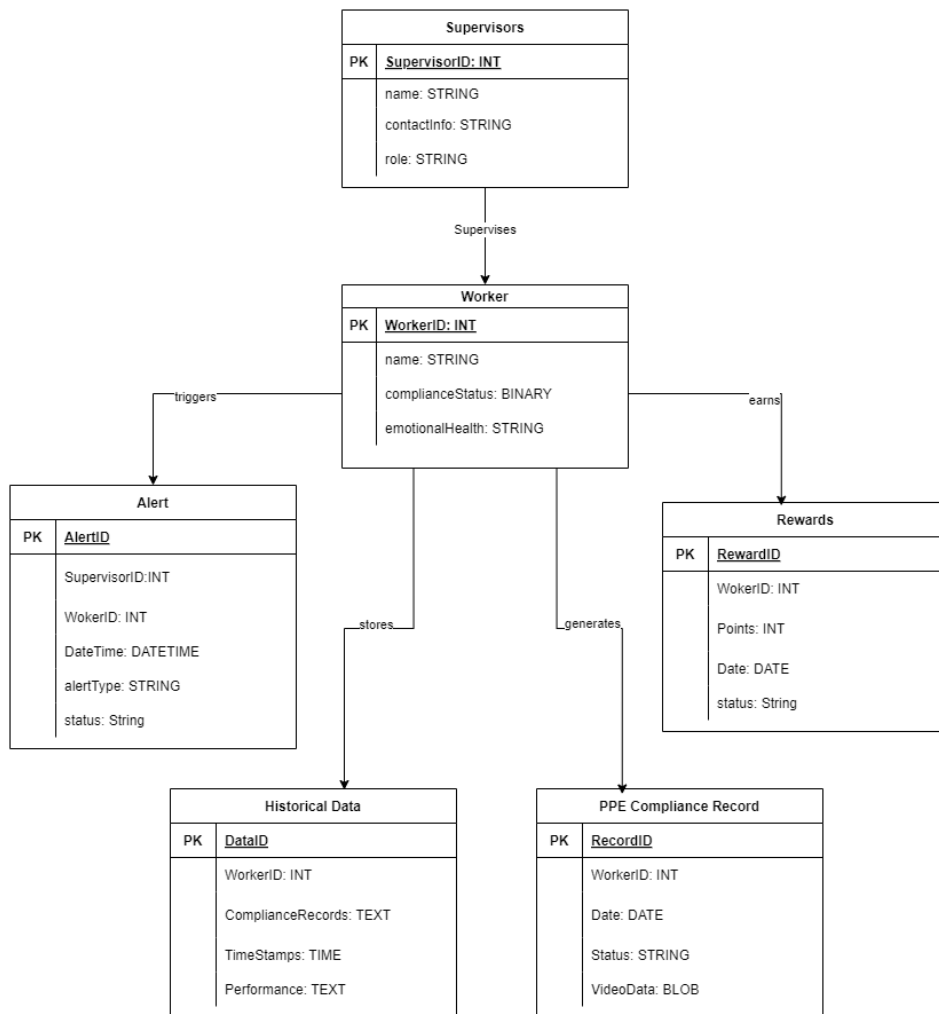


Figure 3.7: JSON schema of UC 01: Detect PPE Compliance

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